INSTITUTIONAL SOFTWARE ENGINEERING SPECIFICATION

ATTENDANCE MANAGEMENT SYSTEM

Project Blueprint & Preliminary Software Requirements Specification

An institutional-grade engineering specification for a centralized web application managing student attendance recording, lecture syllabus delivery tracking, automated 75% threshold alerting, and official Word/A4 document reporting in higher education colleges.

DOCUMENT NATURE
Preliminary Engineering Blueprint & SRS

DOCUMENT VERSION
0.1 — Preliminary Review Draft

ENGINEERING TEAM
Student Development Team (Lead Python Engineers)

TARGET DEPLOYMENT ENVIRONMENT
College / Higher Education Academic Institution

PUBLICATION DATE
September 2026 / Academic Year 2026–2027

PRIMARY STAKEHOLDERS
College Teaching Staff & Academic Administration

1. Executive Summary

The **Attendance Management System (AMS)** is a production-oriented institutional software solution engineered to digitize, standardize, and streamline attendance recording, pedagogical topic delivery logging, regulatory threshold compliance, and official academic documentation across collegiate departments.

Unlike basic academic demonstrations, this platform is conceived as an **institutional-grade management system** capable of supporting multi-department academic structures, enforcing institutional attendance regulations, providing transparent audit trails, and generating official documentation ready for administrative and regulatory use.

> **Status Categorization Discipline:**
> To preserve rigorous engineering integrity during this consultation phase, all specifications are categorized as: **(1) Confirmed Requirements** (mandated by teaching staff), **(2) Proposed Design Decisions** (recommended technical solutions), **(3) Open Questions / TBD** (pending institutional determination), or **(4) Future Extensions** (deferred enhancements).

2. Background and Problem Statement

Higher education institutions consistently experience operational inefficiencies and data integrity risks stemming from manual paper attendance registers and disjointed spreadsheets:

- **Labor-Intensive Record Keeping:** Manual calculation of monthly and cumulative semester percentages consumes valuable faculty time and introduces computational human error.
- **Delayed Remedial Intervention:** Students who fall below the statutory 75% threshold are frequently detected only at the end of the term when it is too late for corrective intervention.
- **Absence of Pedagogical Auditing:** Traditional roll-calls record *who* was present, but fail to maintain a structured institutional record of *what syllabus topics* were delivered during each specific lecture block.
- **Disjointed Administrative Reporting:** Preparing official reports for academic committees, university bodies, parent meetings, and accreditation audits (e.g., NAAC, NBA) requires extensive manual reformatting into A4 and Word-compatible tables.

3. Objectives

The primary engineering and operational objectives derived directly from confirmed requirements are:

1. **Classroom Attendance Recording:** Enable teaching staff to record student attendance efficiently using platform/devices available within the classroom.
2. **Pedagogical Topic Tracking:** Maintain an authoritative institutional record of every conducted lecture, linking class, subject, faculty, timestamps, and specific syllabus points covered.
3. **Real-Time Percentage Computation:** Automatically calculate accurate subject-wise and cumulative semester attendance metrics.
4. **Automated Threshold Warning Dispatch:** Systematically detect students falling below 75% attendance and dispatch automated warning notifications to their official college email.
5. **A4 & Word-Compatible Document Export:** Generate publication-quality monthly and semester reports formatted strictly for ISO A4 printing and Microsoft Word (DOCX) administrative workflows.
6. **Enterprise Data Integrity:** Guarantee transactional reliability, role-based authorization, and tamper-resistant auditability.

4. Scope of the System

In Scope (Confirmed & Proposed Base Platform)	Out of Scope / Not Yet Confirmed (Advanced Scope)
• Multi-tier academic hierarchy (Departments, Semesters, Divisions, Subjects) • Role-Based Access Control (Student, Faculty, Academic Admin) • Lecture session logging with mandatory syllabus topic entries • Atomic attendance capture linked directly to lecture instances • Automated 75% threshold monitoring engine • SMTP email dispatcher for student warning notices • ISO A4-optimized printable PDF and Microsoft Word (.docx) exports • Faculty edit window & administrative audit logging	• Biometric fingerprint / iris hardware integration • Facial recognition / computer vision attendance • GPS / geofencing validation • RFID / NFC smart student ID card readers • Native mobile apps (iOS/Android) — Web-first responsive UI • Automated timetable scheduling & room allocation engines **Advanced mechanisms will only be introduced upon formal institutional request.*

5. Stakeholders and User Roles

The system defines three core institutional user roles with strict boundary controls:

Institutional Role	Proposed Responsibilities	Access Boundaries	Status
Student	• View authenticated personal attendance profile and subject breakdown. • Track syllabus topics covered in conducted lectures. • Receive automated institutional email alerts if attendance falls <75%.	Strictly Read-Only for personal records.	PROPOSED
Teaching Faculty	• Create lecture logs and record topics/points covered. • Record attendance for assigned classes and subjects. • Generate, preview, and download monthly and semester reports. • Submit authorized attendance corrections within permitted timeframes.	Read/Write for assigned classes and courses only.	PROPOSED
Academic Administrator / HOD	• Manage academic master data (departments, classes, courses, enrollments). • Oversee department-wide attendance compliance and analytics. • Audit and authorize post-freeze attendance amendments. • Generate institutional consolidation reports for academic councils.	Departmental or Institutional Global Administrative Access.	PENDING TBD

6. Confirmed Functional Requirements

The following requirements represent the authoritative baseline established by the teaching staff:

Req ID	Requirement Title	Priority	Status	Institutional Reference & Summary
FR-01	Attendance Recording	High	CONFIRMED	Teaching staff must be able to record student attendance using platform/devices available to students. (Capture mechanism TBD).
FR-02	Monthly & Academic Reports	High	CONFIRMED	System must generate monthly summaries and comprehensive semester attendance reports.
FR-03	Lecture & Topic Tracking	High	CONFIRMED	Recording of conducted lectures, date/time duration, and specific syllabus topics/points covered.
FR-04	A4 & Word-Compatible Reporting	High	CONFIRMED	Generated reports must be optimized for A4 paper and easy integration into Microsoft Word (.docx).
FR-05	Attendance % & <75% Email Alerts	High	CONFIRMED	Accurate calculation of subject and semester %; automated warning emails to students below 75%.

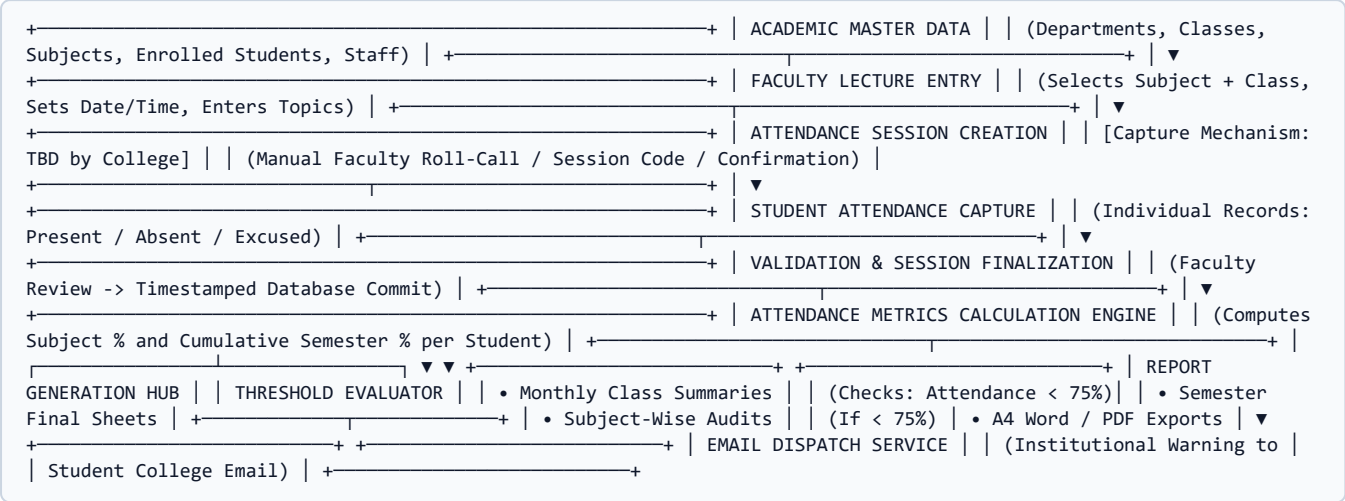
7. Open Questions and Clarifications Required

> **Action Required from Teaching Staff & College Administration:**
> The following items require formal institutional decisions to finalize design and proceed with implementation:

Domain	#	Clarification Question	Options / Engineering Implications	Status
Attendance Capture	Q1	What specific workflow is intended by *"using the platform/device available to students"*?	(A) Faculty manual entry, (B) Faculty-initiated session with student confirmation, (C) Dynamic token entry.	TBD
	Q2	How should proxy attendance be mitigated if student devices are involved?	Short session countdown, one-time code, campus subnet check, or faculty roster sign-off.	TBD
	Q3	What is the policy for retroactive attendance corrections?	Faculty edit window (e.g. 24h) vs. HOD approval required after session locking.	TBD
	Q4	What fallback applies when classroom internet or student device fails?	Faculty manual override portal or offline paper log with batch backfill.	TBD
College Infrastructure	Q5	Does the college have an existing Student Information System (SIS) / ERP?	Standardized CSV/Excel batch upload vs. REST API database sync.	TBD
	Q6	Is there an institutional Single Sign-On (SSO) service available?	Google Workspace / Microsoft 365 OAuth2 vs. standalone Django credentials.	TBD
	Q7	Are student college email addresses assigned and active?	Institutional domain (@college.edu) vs. personal email addresses.	TBD
Academic Rules	Q8	How are cancelled lectures and official holidays handled in calculation?	Excluded from total applicable lectures (denominator).	TBD
	Q9	How are officially sanctioned absences (Medical, Sports, NSS/NCC) treated?	Counted as present, marked excused, or excluded from calculation?	TBD
	Q10	Are practical/lab sessions calculated separately or with weighted credits?	1 Lab = 2 Lecture credits vs. independent practical attendance roster.	TBD
Email Policy	Q11	What is the exact trigger schedule for <75% warning emails?	(A) Immediate on drop, (B) Weekly digest, (C) Monthly checkpoint batch, (D) Manual HOD trigger.	TBD
	Q12	Should warning emails be copied (CC) to Faculty Advisors or HODs?	Direct to student only vs. departmental escalation copy.	TBD
Reporting & Hosting	Q13	Is Word (DOCX), PDF, or both required as official standards?	DOCX for office reformatting; PDF for official signed archives.	TBD
	Q14	Where will the server be hosted (On-premises server vs. Cloud)?	College server room / intranet vs. dedicated Cloud VPS.	TBD

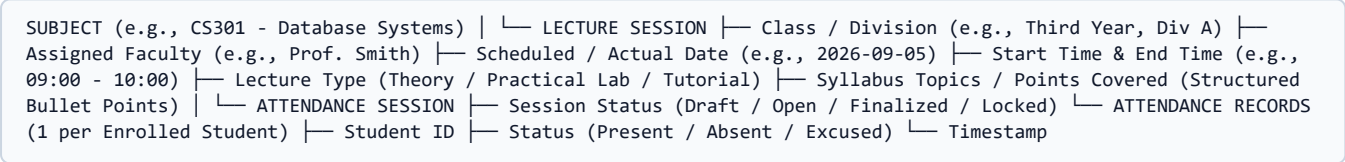
8. Proposed System Workflow

The end-to-end dataflow illustrates how academic records progress from master data creation to attendance capture, threshold checking, and report generation:



9. Proposed Lecture & Topic Tracking Model

In this institutional architecture, the **Lecture is the central operational entity**. Attendance is never recorded as an isolated date-marker; it is directly attached to a specific instructional session:



Institutional Benefits: (1) Fulfills curriculum audit requirements (e.g. NAAC / NBA), (2) Provides student transparency regarding concepts missed during absences, and (3) Ensures instructional continuity for substitute faculty.

10. User Inputs and System Outputs

User Inputs	System-Generated Outputs
Student Inputs: • Login credentials • Session confirmation (if student-side mechanism approved - TBD) **Faculty Inputs:** • Subject, Class, Date, Start/End Time • Detailed syllabus points/topics covered • Present / Absent markers per student roster • Justification for retroactive corrections **Administrator Inputs:** • Master catalogs: Programs, Years, Semesters, Divisions • Faculty & Student profiles, Subject course allocations	**Student Outputs:** • Real-time personal attendance dashboard • Subject-wise attendance % and deficit indicator • Summary log of syllabus points covered • Official <75% warning email notifications **Faculty & Administrator Outputs:** • Live class roster attendance management views • Monthly attendance summary tables • Cumulative semester final attendance sheets • Exportable A4 printable PDFs and editable Word (.docx) documents • Dispatched email audit logs

11. Attendance Calculation Methodology

The system enforces mathematically rigorous attendance tracking:

1. Subject-Wise Attendance Percentage:

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Subject Attendance % = (Present Lectures in Subject / Total Applicable Lectures in Subject) * 100
```

Where Total Applicable Lectures = Total conducted lectures minus officially cancelled lectures.

2. Cumulative Semester Attendance Percentage:

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Semester Attendance % = (Total Present Lectures Across All Subjects / Total Applicable Lectures Across All Subjects) * 100
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3. Statutory 75% Warning Evaluation Rule:

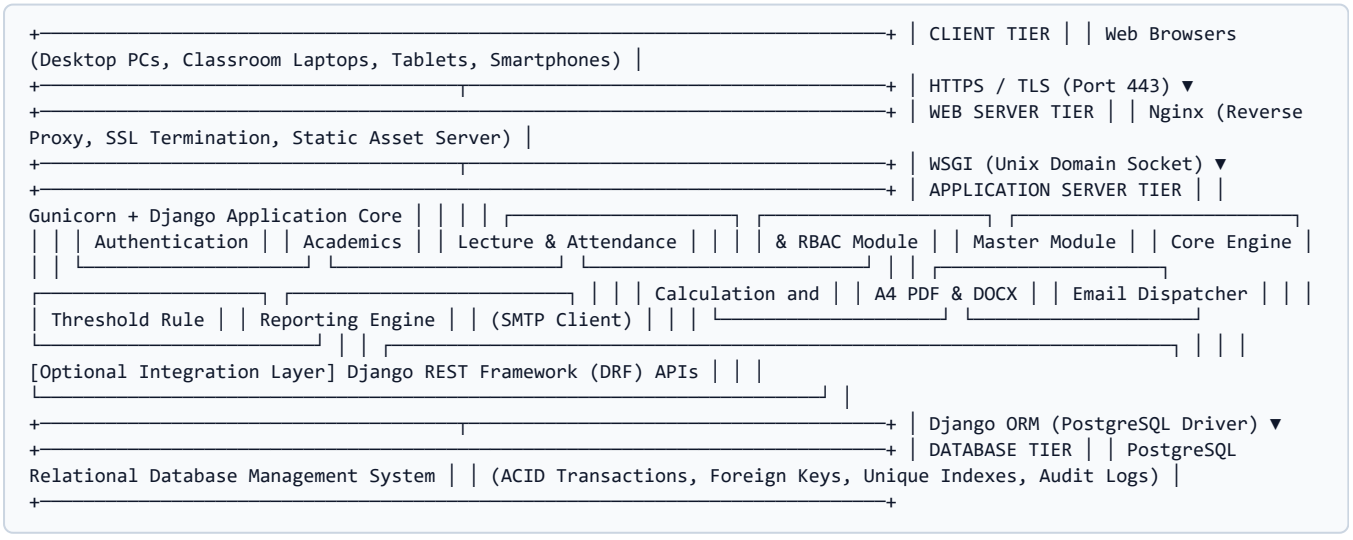
- If **Semester Attendance % < 75.0%**, the student is marked in *Overall Deficit*.
- If **Subject Attendance % < 75.0%** for any enrolled course, the student is marked in *Subject-Specific Deficit*.
- *Trigger:* The notification service drafts and dispatches an official institutional warning email to the student's registered college address.

12. Reporting Design & A4/Word Integration

To satisfy **FR-02** and **FR-04**, the reporting module is decoupled from presentation and generated directly from the authoritative PostgreSQL database:

- **A4 Layout Optimization:** Formatted strictly for standard 210 × 297 mm page dimensions, with 0.5–0.75 in margins, repeating table headers across page breaks, and zero horizontal column overflow.
- **Dual-Format Architecture:**
  - **Microsoft Word (.docx):** Generated using server-side OpenXML formatting, allowing academic departments to incorporate attendance tables directly into administrative notices and formal college templates.
  - **Printable PDF (.pdf):** Tamper-resistant, publication-quality format containing institutional headers, department seals, and signature blocks (*Subject Teacher*, *Class Coordinator*, *Head of Department*).
- **Standard Report Types:** (1) Monthly Class Attendance Sheets, (2) Cumulative Semester Attendance Sheets, (3) Subject-Specific Audit Sheets, and (4) Below-75% Deficit Summaries.

13. Proposed System Architecture



14. Preliminary Conceptual Data Model

The relational database schema is structured around normalized institutional entities:

- **Department & Program:** Academic hierarchy and degree structures.
- **AcademicYear & Semester:** Temporal academic terms (e.g. 2026–2027, Sem V).
- **ClassDivision & Subject:** Student cohorts and course catalog mappings.
- **Faculty & Student:** User identity profiles linked to system authentication.
- **Enrollment:** Maps students to specific class divisions and elective subjects.
- **Lecture:** Conducted session (Date, Time, Faculty, Class, Syllabus Topics Covered).
- **AttendanceSession & Record:** Operational session linking individual atomic student statuses (`PRESENT`, `ABSENT`, `EXCUSED`).
- **NotificationLog & AuditLog:** Dispatched email audit trail and security modification log.

15. Security, Authorization & Auditability

- **Principle of Least Privilege (RBAC):** Students access solely their personal records; faculty access assigned classes; administrators hold audited institutional access.
- **Session Security & Defense-in-Depth:** HTTPS/TLS encryption, CSRF protection on all forms, SQL injection defense via Django ORM, and HttpOnly session cookies.
- **Immutability & Correction Logs:** Attendance sessions freeze after an authorized window. All post-freeze modifications require justification and generate immutable audit logs (`user_id`, `timestamp`, `previous_value`, `new_value`).

16. Non-Functional Requirements (NFRs)

Quality Attribute	Institutional Engineering Specification
Reliability	ACID transactions on all batch attendance submissions to prevent partial or corrupted records.
Performance	Server response time < 1.5 seconds under standard load. Concurrent entry by 50+ faculty simultaneously.
Usability	Classroom attendance entry workflow completed in < 60 seconds for a roster of 60 students.
Availability	99.5% uptime during operational college hours (08:00 to 18:00, Monday through Saturday).
Data Retention	Historical attendance and syllabus topic logs preserved for statutory minimum of 5 years.

17. Deployment & Technology Stack

Component	Proposed Technology	Technical Rationale
Backend	Python 3.11+ / Django 5.x	Industry standard for institutional systems. Built-in ORM, robust RBAC, automatic CSRF/XSS protection, mature administrative dashboard.
Database	PostgreSQL 16	Enterprise-grade relational database with high concurrent transaction handling, data integrity constraints, and rock-solid reliability.
Frontend UI	Django Templates + Bootstrap 5 + Vanilla JS	Lightning-fast page loads, full mobile/tablet responsiveness, zero client-side build complexity, easy long-term institutional maintainability.
API Layer	Django REST Framework (DRF)	Provides modular endpoints for future integrations (e.g. mobile app, college ERP) without altering core models.
Documents	python-docx & WeasyPrint / ReportLab	Server-side compiled A4 PDFs and native Microsoft Word (.docx) document generation.
Hosting	Linux (Ubuntu 24.04), Nginx, Gunicorn	Proven production stack for high-availability web applications with reverse proxying and automated systemd management.

18. Risks & Mitigation Strategies

Identified Risk	Severity	Proposed Technical & Operational Mitigation
Proxy Attendance	Medium	Short session timer, faculty roster verification before final locking, or faculty-device-only entry.
Classroom Network Failure	High	Client-side offline caching (LocalStorage) or fallback paper roster with batch backfill.
Clerical Absence Errors	Medium	Authorized 24-hour faculty correction window with mandatory change audit logging.
Email Delivery Failures	Medium	Persistent `NotificationLog` table tracking SMTP delivery status codes and timestamps.
Data Loss / Hardware Failure	High	Automated daily PostgreSQL `pg_dump` backups archived to secure secondary/cloud storage.

19. Decisions Pending Approval & Development Roadmap

Item #	Strategic Decision	Status	Recommended Option for Review
D-01	Attendance Capture Workflow	TBD	Faculty-controlled interface with optional student verification session.
D-02	Master Data Import	TBD	Standardized CSV / Excel batch import template.
D-03	Email Notification Trigger	TBD	Weekly automated digest + immediate alert upon dropping below 75%.
D-04	Official Report Standard	TBD	Dual Export: Microsoft Word (.docx) + Signed A4 PDF (.pdf).
D-05	Excused Absence Handling	TBD	Formal `EXCUSED` status excluded from absence penalty upon medical proof.

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[Phase 1: Requirements & Blueprint Sign-Off] ──> [Phase 2: Database & Security Architecture] │ ▼ [Phase 4: Reports, Calculation & Email Engine] <── [Phase 3: Core Attendance & Lecture Module] │ ▼ [Phase 5: Testing & Departmental Pilot Run] ──> [Phase 6: Full Institutional Deployment]
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